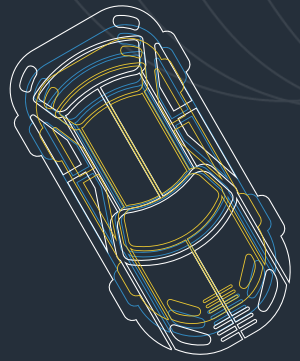
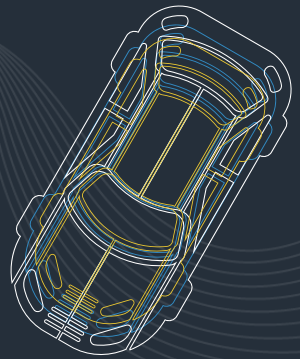


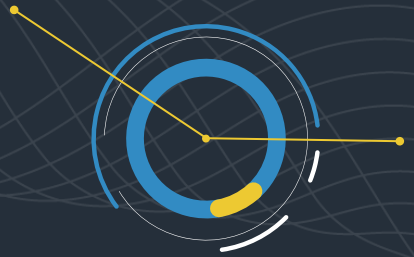


Hydrogen Car Renewal

Justin Chai and Ben Horvat
Advisor: Prof. Everbach
Swarthmore College E90
Spring 2024



Presentation Outline



01

Shell Eco-Marathon/ Previous Projects

02

Existing Conditions/ Motivation

03

Relevant Theory

04

Progress

05

Struggles

06

Future Plans

Shell Eco-Marathon

- **Worldwide Engineering Competition** for Energy Efficiency and Low Carbon Transportation Solutions!
  - NA: Indianapolis Motor Speedway
    - Like the less fast innovation F1
- **Multiple Categories:** Prototype and Urban
- **Different Powertrain Systems: Hydrogen**, Combustion, Battery EVs





Technical Inspections

Before they get on track, Prototype vehicles need to pass a series of 11 technical inspections. For some, the only goal is getting through this test. Many never even make it that far.



Not yet started 1. Inspection Queue Entry	0 / 3
Not yet started 2. Driver's Control	0 / 12
Not yet started 3. Vehicle Weight	0 / 2
Not yet started 4. Brakes	0 / 9
Not yet started 5. Vehicle Dimensions	0 / 6
Not yet started 6. Seat Belts, Roll bar & Exit	0 / 12
Not yet started 7. Visibility	0 / 7
Not yet started 8. Mechanical Vehicle Design	0 / 20
Not yet started 9. Electrical Vehicle Design	0 / 31
Not yet started 10. Energy Verification	0 / 17
Not yet started 11. Final Check	0 / 4

Technical Inspections

BEFORE RACING, need to pass the 11 rounds of testing -> **lots of design constraints**:

- Turning radius
- Dimensions of car
- Hydrogen FC integration
- Braking tests at different intensity
- Safety/Visibility of driver

Focused Design Constraints

Size Regulation

-

Safety Regulation

- Front and back suspension
- All hydraulic brakes
- Firewall between driver and engine
- Proper electrical harnessing

Performance Regulation

- Turning Radius of
- Passing __ degree brake test

Prior Vehicles and Past Troubles


Current Status


2018 Front Suspension

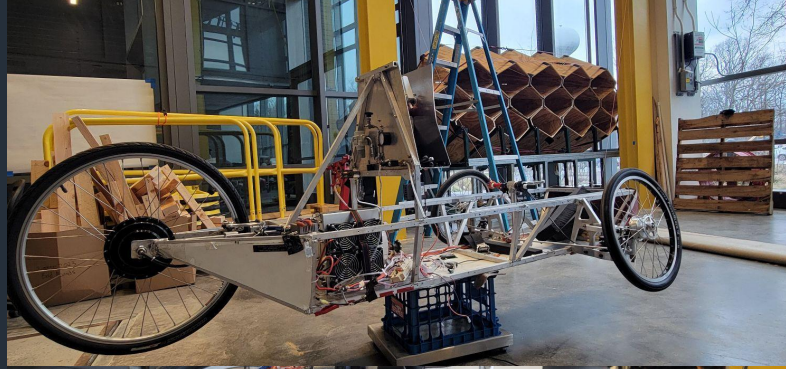

2012 - Original Chassis Construction

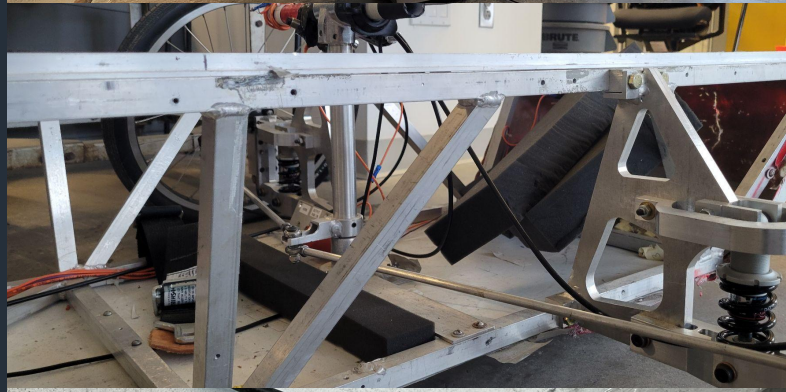


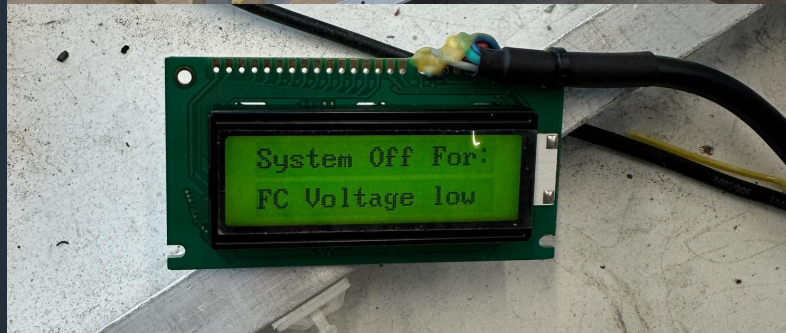
2015 Competition Team

Motivations for an Overhaul

Outdated Chassis Model	❖ Current chassis was made in ‘13 ❖ No aerodynamic considerations ❖ New size requirements
Hydraulic Brakes	❖ Hydraulic disk-pad brakes are standardized and required
Outdated Materials	❖ Aluminum chassis adds unnecessary weight and is restricting performance
New front and rear suspension	❖ Front suspension is hefty and complicates steering ❖ No rear suspension system
Steering and Driver Safety Systems	❖ Steering columns need to be redesigned for turning radius requirements ❖ Safety electronics need an update
Fuel Cell Woes	❖ Current FC is inconsistent and needs to be assessed for further use







FROM CURRENT





01

Work cut out for us?





To New Standard

Main Goals:

How do we get more efficient?

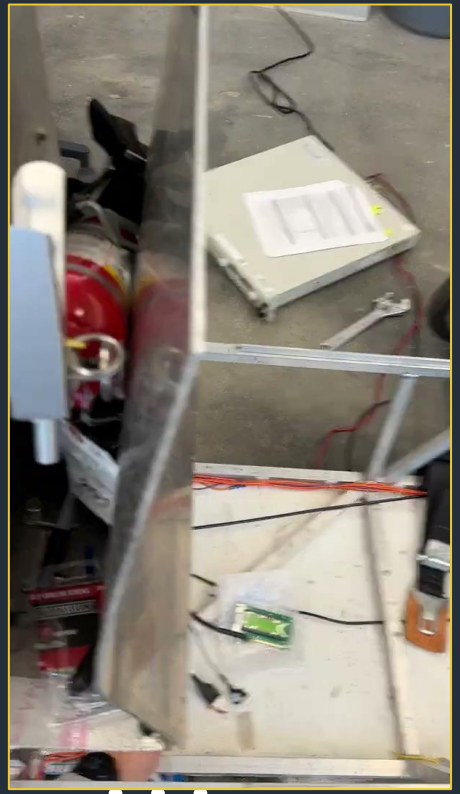
Fuel Cell Renewal & Feasibility Assessment

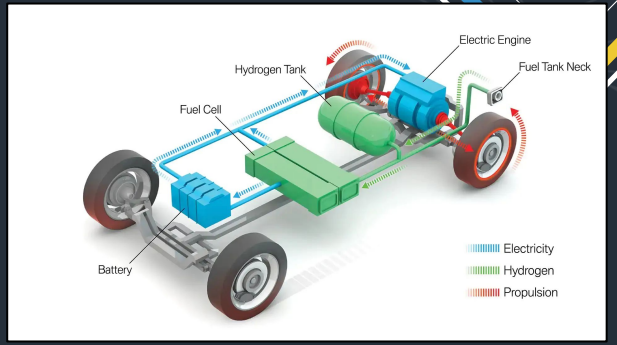
How does a HFCV work?

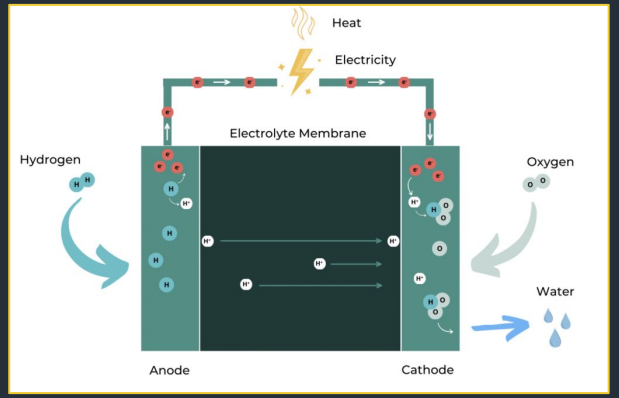
- H_2 gas moves along a membrane where it merges with O_2 gas to make water and electricity
- An electric motor powers the vehicle -> 0 emission kinematic propulsion!

Current FC woes

- FC was bought in ‘19 and has had multiple start-up issues
- Possible need for FC rehydration & system renewals
- No benchmarking of potential power







Fuel Efficiency is King!

Shell Eco-Marathon measures the amount of hydrogen used to drive **3.9kms in less than 35 mins**.

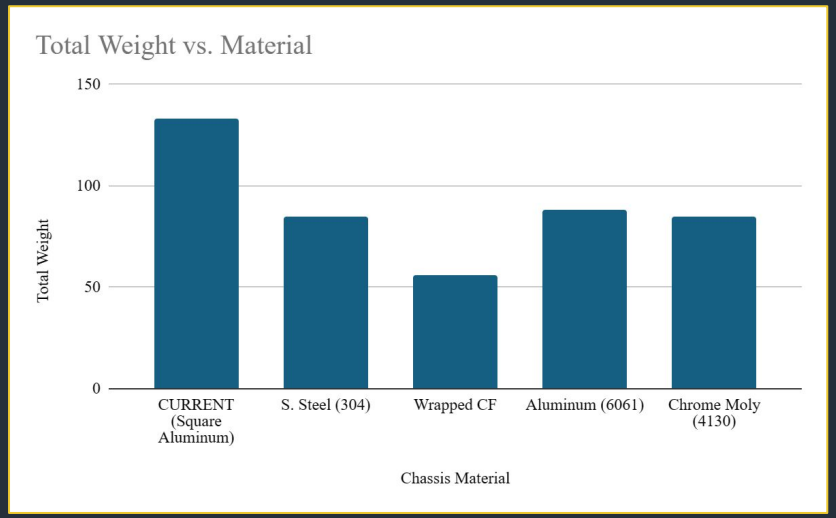
Weight kills efficiency

Our fuel efficiency model estimates **~3-4 MPGe gained per pound reduced**.

240+ MPG Gained

A **potential >20% fuel efficiency increase** by swapping to lighter material!

Material Impacts



CF Tube Δ

Overall Weight (Including Driver):

243 ➔ 166 lbs (31% Decrease)

Fuel Efficiency:

1288➔ 1598MPGe (24% Increase)

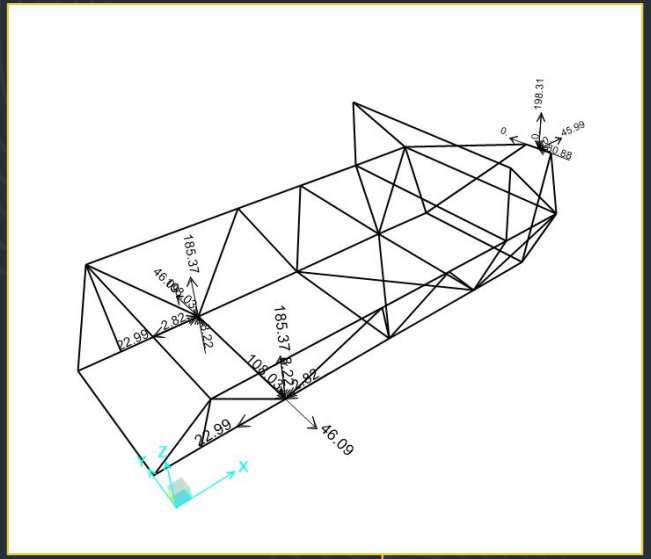
Suspension

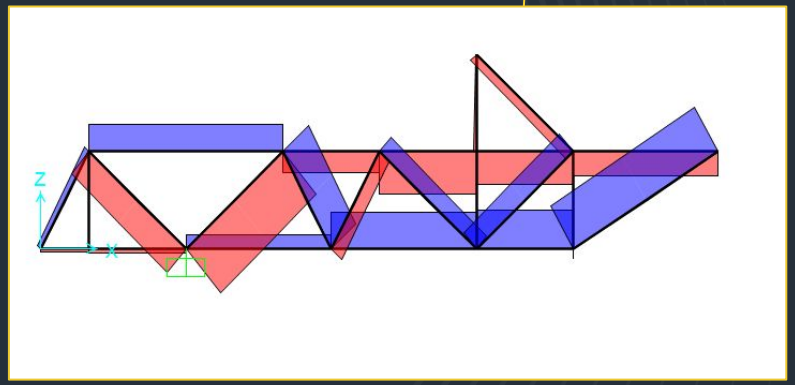
Overall Weight:

25 ➔ ~12 lbs (With Added Back)

2:

New Chassis & Materials Research





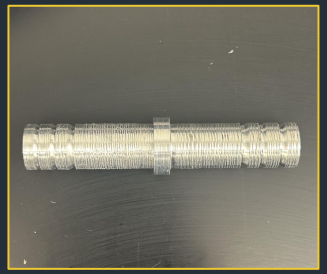
Connection Methods: Carbon Fiber Tubes

Aluminum Inserts

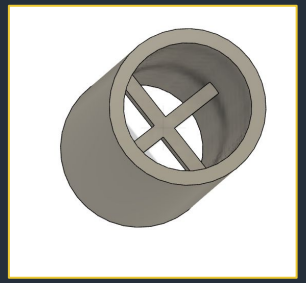
Benefits:
- Reliable material strength
- Machining Constraints

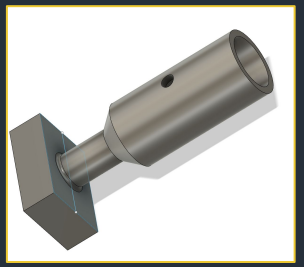
Drawbacks:
- Heavier
- Welding is difficult

Potential Additional Strength:
- Cable/String through rod









3D Printed Collars

Benefits:
- Lightweight and strong
- Flexible in design

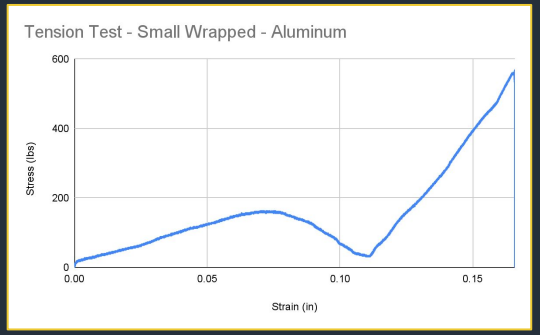
Drawbacks:
- Layer Shear Failure
- Misprints

Potential Additional Strength:
- Screw Connections

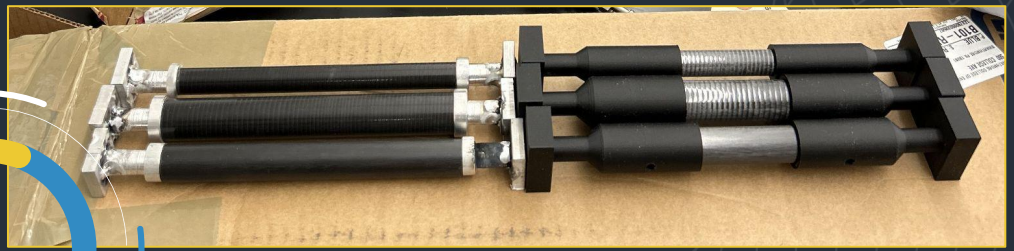
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Tension Tests









Tension Test Results + Cost Analysis

Maximum Simulated Loads (200 lb Driver): 300 lbs (T) & 540 lbs (C)
Design Goals: 1000 lbs (T) & 2000 lbs (C)

	SW (.5” D)	LW (.75” D)	Average (Wrapped)
Maximum (lbs)	561.1629	596.35785	578.760375
Modulus (lbs/in)	9425	9297.626316	9361.313158
Est. Displacement before Fail	0.05	0.055	0.055

Failures so far

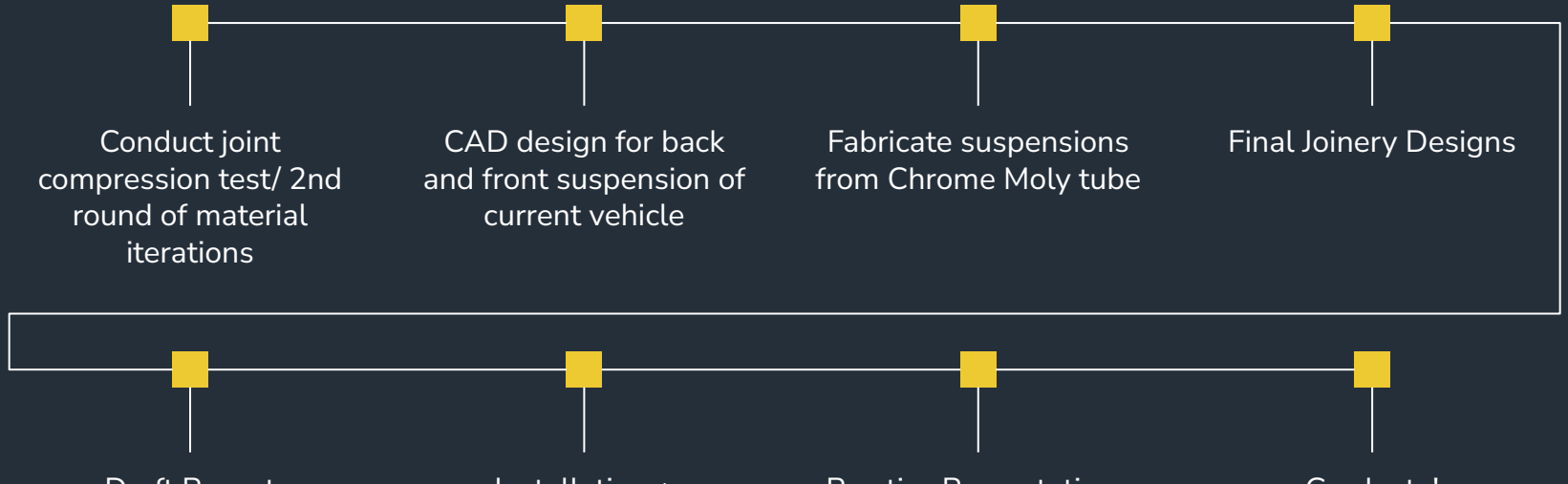


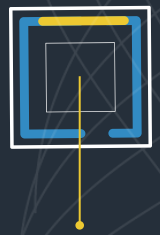






Timeline and Next Steps





Thanks!

Do you have any questions?

